CCNA Scenario Based Questions

Networking troubleshooting interview practice: Q1 to Q8

Simple, step-by-step answers you can say in an interview.

Highlighted words are the key terms to remember. Each answer starts with a blue question bar, then numbered steps, then a closing line.

Q1. PC1 cannot ping PC2. How would you troubleshoot?

"I'd troubleshoot step by step to find where communication is failing.

1. **Check the physical connection:** I'd confirm both PCs are powered on, cables are connected, and network interfaces and switch ports are up.

2. **Verify IP settings:** I'd check both PCs' IP addresses and subnet masks, confirm the destination IP is correct, and check their default gateways if they are on different networks.

3. **Check the network path:**

   - **Same subnet:** I'd check VLAN membership, ARP entries, switch ports and links between switches.
   - **Different subnets:** I'd ping PC1's default gateway, use tracert/traceroute to help locate the failure, and check routes in both directions, including PC2's gateway. A missing traceroute reply alone doesn't prove that device is down.

4. **Check security rules:** I'd inspect firewalls, ACLs and logs for blocked ICMP traffic. With approval, I'd temporarily allow ICMP specifically between these PCs to test, then remove the test rule. I'd avoid disabling the whole firewall or ACL. In an isolated lab, a brief firewall-disable test is possible, but I'd re-enable it immediately.

5. **Consider restarting:** If needed after the checks, I'd save work and restart the affected PC. I wouldn't blindly reboot a shared switch or router because it affects other users; that needs approval and a planned interruption.

Finally, I'd fix the identified cause and ping again from both PCs to confirm communication works. A failed ping alone doesn't mean all network connectivity is down."

Q2. PC1 can ping PC2, but traceroute fails. How would you troubleshoot?

"Since ping works, PC2 is reachable. I'd check why traceroute isn't getting the replies it needs.

1. **Confirm the target:** I'd make sure ping and traceroute use the same destination IP. Then check whether traceroute shows an error, some stars (*), or never reaches PC2.

2. **Check the method:** Windows tracert uses ICMP echo test packets. Many traceroute tools use UDP by default, so the traffic may be treated differently from ping.

3. **Check firewall rules and ACLs:** They may allow ping but block traceroute's test packets or the ICMP Time Exceeded replies that routers send to identify themselves. Routers may also limit these replies.

4. **Test further:** If supported, I'd try ICMP traceroute instead of UDP and compare from another PC. If needed, I'd use Wireshark or tcpdump to check which test packets leave and which replies return.

5. **Fix and verify:** If I find a blocking rule, I'd request an approved, specific change and test again. I wouldn't disable the entire firewall or ACL.

Stars mean no reply was received for those probes. They don't necessarily mean the router is down, because it may still forward traffic normally. I'd also check whether the actual application works."

Q3. You cannot fetch data from a server. How would you troubleshoot?

"I'd first find out whether the problem is the connection, the server's application, or my access permissions.

1. **Understand the issue:** I'd check the exact error, which application or resource I'm accessing, and whether it affects only me or other users too.

2. **Check the connection:** I'd verify my PC's network connection, IP settings and the server address. I'd ping the server, but remember that a failed ping may simply mean ping is blocked.

3. **Check name resolution:** I'd use nslookup to check whether the server name resolves to the expected IP address. DNS is what translates a server name into an IP address.

4. **Test the required service:** A server can reply to ping while its application isn't working. I'd test the application's required port, its numbered network door. For example, for an HTTPS service on Windows:

   ```
   Test-NetConnection 192.168.1.10 -Port 443
   ```

   If that test fails, I'd investigate the service, routing, firewall and ACL rules rather than assume the server is down.

5. **Check access and the request:** If I can connect but get "access denied", I'd check my login and permissions. I'd also confirm the requested file, data or resource exists and the path is correct.

6. **Check the server:** With access or help from the server team, I'd check that the service is running, review its logs, and check disk space, CPU, memory and any backend it depends on.

Finally, I'd fix the identified cause, make only approved changes, and confirm I can fetch the actual data. Ping working alone isn't enough."

Q4. You are having frequent disconnections from a server. How would you troubleshoot?

"First, I'd check whether only my PC is disconnecting or other users are affected too. I'd also check whether it happens with one server or all servers.

1. **Check the connection:** inspect cables, ports and Wi-Fi, then continuously ping the gateway and server to see whether missed replies happen at the same time as the disconnection. On Windows: `ping -t <server-IP>`. Missing replies are a clue, not proof, because ICMP replies can be blocked or limited.

2. **Narrow down the issue:** test from another PC and try another server. This helps identify whether the issue is with my PC, the network path or that server.

3. **Check security settings:** review firewall/ACL rules and session timeouts to see whether connections are being blocked or closed. A fixed idle period before the drop points to a timeout setting.

4. **Check device and server health:** look for port errors, links going up and down, high CPU or memory usage, and services restarting. Compare logs with the exact time of the disconnections.

Finally, I'd fix the cause found and monitor the connection to confirm it stays stable. I wouldn't blindly reboot devices or disable security."

Easy order to remember: Connection → Narrow it down → Rules → Health and logs → Retest.

Q5. You have an IPsec tunnel but cannot access resources through it. How would you troubleshoot?

"A tunnel showing "up" doesn't guarantee that my traffic can pass through it. I'd check these steps:

1. **Check tunnel status:** I'd confirm both gateways have an active IPsec connection for the networks I'm trying to connect. If it isn't established, I'd compare the VPN settings on both sides and check the error logs.
2. **Check routes:** My side needs a route to the remote network through the VPN, and the remote side needs a route back to my network.
3. **Check the allowed networks:** I'd confirm my PC's network and the server's network are included in the VPN's traffic selectors. These settings define which traffic the VPN protects.
4. **Check firewall and NAT rules:** I'd confirm the required traffic is allowed on both sides, and that address translation isn't changing the traffic in an unintended way.
5. **Test the resource:** I'd check whether the server is reachable by IP and whether its required service works. I'd check DNS separately to confirm the server name points to the correct IP.
6. **Check counters and logs:** While testing, I'd watch the tunnel's sent and received packet counts on both gateways. This helps narrow down whether traffic enters the VPN, reaches the other side and gets a reply.

Finally, I'd fix the identified cause with approval and test access to the actual resource again."

Remember: Tunnel → Routes → Allowed networks → Firewall/NAT → Resource → Logs.

Q6. R1 → R2 → R3. OSPF runs on R1 and R3, but not R2. Will R1 and R3 become OSPF neighbors?

"No, not in this topology.

1. OSPF routers use Hello packets to discover neighboring routers on their link.
2. R1's normal OSPF Hellos don't travel through R2 to R3. R2 doesn't forward those local Hello packets, and it isn't running OSPF itself.
3. So R1 and R3 cannot discover each other or form an OSPF neighborship.

To connect these networks using OSPF, I'd enable OSPF on R2's relevant interfaces too, with matching neighbor settings on each link. Then R1 becomes R2's neighbor, and R2 becomes R3's neighbor.

R1 and R3 can learn routes through R2, but they still aren't direct OSPF neighbors."

Remember: learning a route through another router is different from being its direct OSPF neighbor.

Q7. Where are NAT-D and NAT-T used in IPsec?

Think of a VPN gateway sitting behind an internet router that changes its private IP into a public IP. That address change is NAT (Network Address Translation).

1. **NAT-D = NAT Discovery.** During VPN setup, the two VPN endpoints check whether an IP address or port was changed between them. NAT-D detects that change.

2. **NAT-T = NAT Traversal.** If NAT is detected, NAT-T helps the VPN traffic cross it. It puts the encrypted IPsec data, called ESP, inside UDP packets, normally using port 4500. The NAT device can then track the traffic using UDP ports.

3. **Usual flow:** Start VPN setup on UDP 500 → detect NAT → use UDP 4500 for IKE setup traffic and UDP-wrapped ESP data.

4. **Simple example:** The VPN gateway is 192.168.1.10. The internet router translates it to a public IP. NAT-D detects the translation. NAT-T adds the UDP wrapping so the encrypted traffic can pass through. The remote VPN endpoint removes that wrapping and decrypts the data.

NAT-T doesn't perform the address translation itself. In tunnel mode, the NAT device changes the outside addressing, not the original addresses hidden inside the encrypted packet.

Interview line: "NAT-D detects whether NAT exists between VPN endpoints. NAT-T allows IPsec traffic to pass through NAT by wrapping ESP in UDP, normally on port 4500."

For the message numbers on the IKEv1 Main Mode board: messages 1-2 advertise NAT-T support, 3-4 detect NAT, and 5-6 move to UDP 4500 if NAT is detected. That sequence is specifically IKEv1 Main Mode, not every IPsec setup.

Q8. You cannot access a server at Site B from Site A through an IPsec tunnel. How would you troubleshoot?

"I'd check the connection step by step, from my PC to the server.

1. **Basic connection:** I'd check my PC's IP settings and connection to its gateway. I'd also confirm someone at Site B can access the server locally.
2. **Tunnel status:** I'd confirm the IPsec tunnel is established for the required networks on both gateways. If not, I'd check the VPN settings and error logs.
3. **Routes:** I'd check that Site A sends traffic for Site B through the VPN, and Site B has a route back to Site A.
4. **Rules:** I'd check that both networks are included in the VPN's allowed traffic, that firewall/ACL rules allow the required service, and that NAT isn't changing addresses unexpectedly.
5. **Server and service:** I'd check the server's IP, default gateway, local firewall and whether its application is running. I'd check DNS if access by name fails.
6. **Find where traffic stops:** While testing, I'd compare VPN packet counters and logs on both sides. If needed, I'd capture packets to see where requests or replies disappear.

Finally, I'd fix the identified cause and test the actual application again."

This is the same core problem as Q5, now shown as Site A → VPN → Site B → Server. A tunnel being up doesn't guarantee the server is accessible.

Reference sources

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